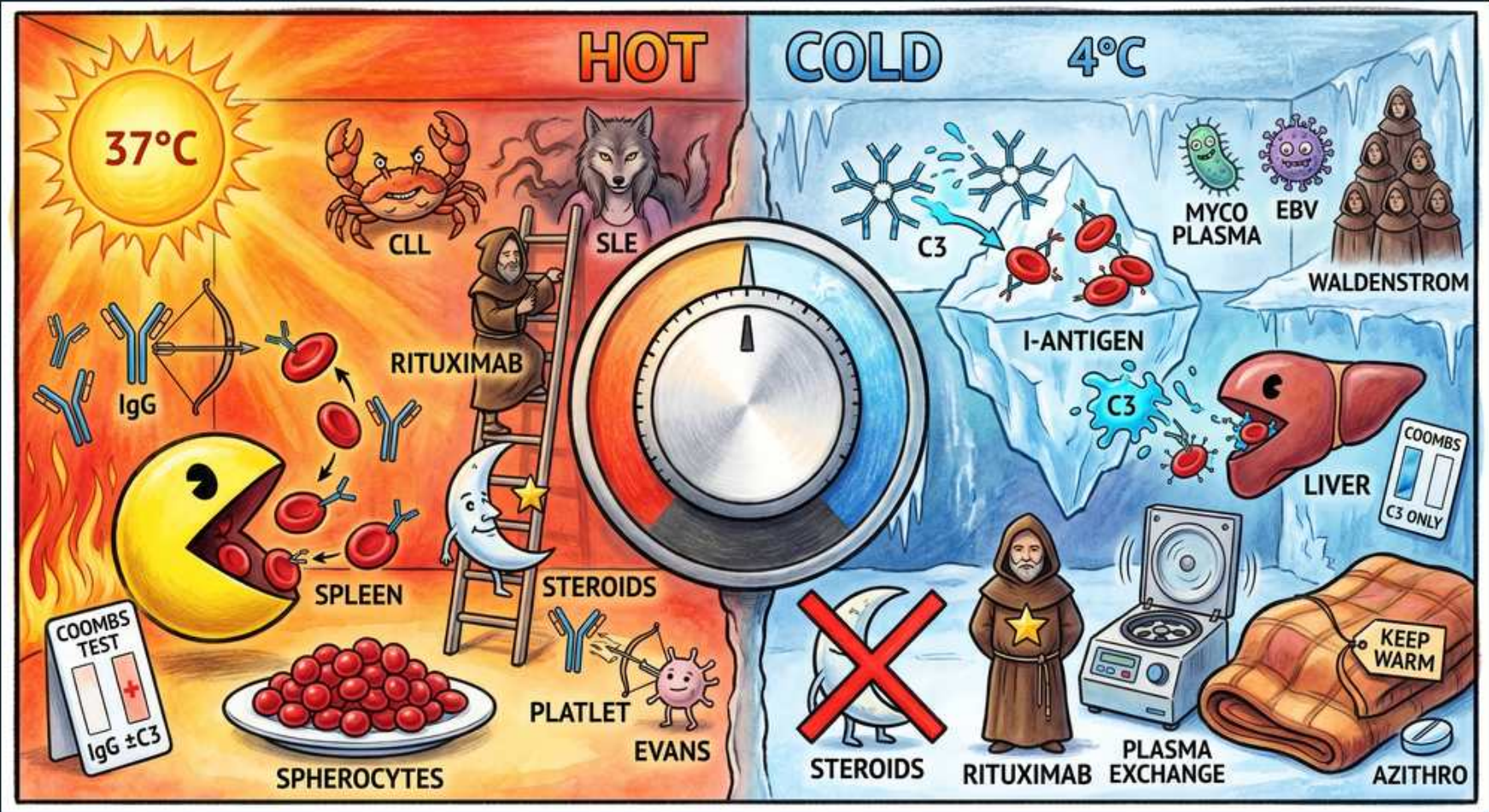


Autoimmune Hemolytic Anemia — Warm vs Cold



1. Warm AIHA — 37°C

WHAT YOU SEE

A blazing sun stamped with 37°C anchors the HOT left half — body temperature, where warm-antibody hemolysis runs hottest.

WHAT IT ENCODES

Warm autoimmune hemolytic anemia is the most common AIHA; the autoantibody is IgG binding maximally at body temperature (37°C).

BOARD TRAP

Calling it IgM/cold — warm AIHA is IgG at 37°C.

2. IgG antibodies

WHAT YOU SEE

Y-shaped IgG antibodies float labeled 'IgG' on the warm side, latching onto a red cell — the warm autoantibody coating its target.

WHAT IT ENCODES

IgG-coated red cells are partially phagocytosed by splenic macrophages, producing spherocytes; the antibody is often anti-Rh (anti-e).

BOARD TRAP

Expecting complement-mediated intravascular lysis — warm AIHA is mainly extravascular/splenic.

3. Spleen clearance

WHAT YOU SEE

A hungry yellow Pac-Man chomps down a line of red cells labeled 'SPLEEN' — the splenic macrophage devouring antibody-coated erythrocytes.

WHAT IT ENCODES

The spleen is the site of clearance in warm AIHA and splenomegaly is common.

BOARD TRAP

Liver clearance — that is cold agglutinin disease, not warm.

4. Spherocytes

WHAT YOU SEE

A heaped pile of small, round, hyperdense red spheres labeled 'SPHEROCYTES' along the bottom-left — cells left behind after the spleen nibbles membrane away.

WHAT IT ENCODES

Membrane nibbling by splenic macrophages produces spherocytes; warm AIHA is spherocytic like HS but the DAT is positive.

BOARD TRAP

Reading spherocytes as hereditary spherocytosis — the DAT separates them.

5. Coombs IgG ±C3

WHAT YOU SEE

A Coombs test card reading 'IgG ±C3' sits in the lower-left corner — the direct antiglobulin test that pins down warm disease.

WHAT IT ENCODES

A DAT positive for IgG (with or without C3) defines warm AIHA.

BOARD TRAP

A C3-only DAT — that is COLD agglutinin disease, not warm.

6. Steroids — first line

WHAT YOU SEE

A syringe labeled 'STEROIDS' stands upright on the warm side — corticosteroids, the first move in warm AIHA.

WHAT IT ENCODES

First-line therapy for warm AIHA is corticosteroids (prednisone ~1 mg/kg), plus folate supplementation.

BOARD TRAP

Using steroids first-line in COLD AIHA — they do not work in cold disease.

7. Rituximab

WHAT YOU SEE

A hooded archer-knight climbs a ladder labeled 'RITUXIMAB' on the warm side — the second-line escalation.

WHAT IT ENCODES

Rituximab is second-line for steroid-refractory or relapsed warm AIHA; splenectomy is another option.

8. CLL / SLE (secondary)

WHAT YOU SEE

A crab labeled 'CLL' and a snarling wolf labeled 'SLE' prowl the top of the warm side — the secondary drivers lurking behind warm hemolysis.

WHAT IT ENCODES

Secondary warm AIHA is driven by CLL, lymphoma, SLE or drugs — evaluate for an underlying cause.

9. Evans syndrome

WHAT YOU SEE

A spiky virus-faced platelet labeled 'EVANS' sits beside a labeled platelet at the warm-side base — warm AIHA paired with immune thrombocytopenia.

WHAT IT ENCODES

Evans syndrome = warm AIHA combined with immune thrombocytopenia (and sometimes immune neutropenia).

10. Cold AIHA — IgM, 4°C

WHAT YOU SEE

An icy '4°C' banner crowns the frozen COLD right half — the chilled peripheral temperature where cold agglutinins clump.

WHAT IT ENCODES

Cold agglutinin disease is an IgM autoantibody binding best in the cold (0–4°C), agglutinating red cells in cool peripheral blood and causing acrocyanosis.

BOARD TRAP

Calling it IgG — cold disease is IgM.

11. I antigen (anti-I)

WHAT YOU SEE

A cluster of clumped red cells labeled 'I-ANTIGEN' sits on the iceberg — agglutinated erythrocytes targeted by anti-I.

WHAT IT ENCODES

Cold agglutinins target the I antigen (anti-I).

BOARD TRAP

Anti-i is associated with infectious mononucleosis (EBV), whereas anti-I is Mycoplasma/idiopathic — do not swap them.

12. C3 only — liver clearance

WHAT YOU SEE

A large reddish-brown liver labeled 'LIVER' on the cold side, with a 'COOMBS C3 ONLY' card beside it — complement-coated cells cleared by hepatic Kupffer cells.

WHAT IT ENCODES

IgM fixes complement then falls off, so red cells are coated only with C3 and are cleared by the LIVER (Kupffer cells); the DAT is C3-only (IgG negative).

BOARD TRAP

Expecting splenic clearance or an IgG DAT — cold disease is hepatic clearance with a C3-only DAT.

13. Mycoplasma / EBV / Waldenström

WHAT YOU SEE

Three microbe characters across the top-right — a green 'MYCO PLASMA' bug, a purple 'EBV' virus, and a stacked 'WALDENSTROM' tower of cells.

WHAT IT ENCODES

Secondary cold disease: Mycoplasma pneumoniae (anti-I) and EBV (anti-i) post-infectious; chronic disease with Waldenström macroglobulinemia / lymphoma (monoclonal IgM).

14. Cold treatment — keep warm, rituximab, plasma exchange

WHAT YOU SEE

A cozy folded 'KEEP WARM' blanket, an 'AZITHRO' label, a hooded monk (rituximab) and a centrifuge marked 'PLASMA EXCHANGE' line the cold-side base — while a big red X strikes through the steroids syringe.

WHAT IT ENCODES

Manage cold AIHA by keeping the patient WARM, treating the underlying cause (azithromycin for Mycoplasma), and giving rituximab for primary disease; plasma exchange helps acute severe cases because IgM is intravascular.

BOARD TRAP

Treating cold AIHA with steroids or splenectomy — both are ineffective because clearance is hepatic/complement-mediated; use rituximab (± the complement inhibitor sutimlimab).
